

Task – 5 (Week – 1):

Paper	Year	Key Idea	Relevance
A Quantum Approximate Optimization Algorithm <i>(Farhi, Goldstone, Gutmann)</i>	2014	The Genesis of QAOA: Proves a shallow-depth quantum circuit can approximate combinatorial optimization solutions. It alternates between a Problem Hamiltonian (H_C) to encode the cost function and a Mixer Hamiltonian (H_B) to drive transitions out of local minima.	Serves as the absolute mathematical baseline for understanding how QAOA operates and why parameterized gates are used.
Quantum Computing for Finance: Overview and Prospects <i>(Orús, Múgel, Lizaso)</i>	2019	The Industry Taxonomy: Categorizes financial problems that map to quantum systems (Optimization, Machine Learning, Simulation). Details the structural differences and timelines between Quantum Annealing and Gate-Model computers.	Establishes the business case for why financial institutions are investing in quantum hardware and identifies QUBO as the near-term target.
Solving the Optimal Trading Trajectory Problem Using a Quantum Annealer <i>(Rosenberg et al.)</i>	2016	Dynamic Multi-Period Mapping: Pioneers the translation of a trading trajectory (balancing returns, risk, and transaction costs over consecutive days) into a QUBO format digestible	Essential for understanding how to turn dynamic financial constraints (dollars and risk) into strict binary math (0 and 1).

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		by quantum hardware.	
Quantum Optimization for Portfolio Management <i>(Egger et al.)</i>	2020	Markowitz on Quantum: Compares classical solvers against QAOA and VQE for Markowitz Portfolio Selection. Explores applying massive mathematical penalties to enforce asset limits and details how hardware noise flattens the energy landscape.	Directly bridges the gap between pure quantum theory and applied Modern Portfolio Theory (MPT).
Warm-Starting Quantum Optimization <i>(Egger, Marecek, Woerner)</i>	2021	Beating Barren Plateaus: Proposes using a classical computer to find an approximate solution first, and injecting that guess into the starting state of the QAOA circuit. This drastically reduces the required depth (p) of the quantum circuit.	Highly relevant for Noisy Intermediate-Scale Quantum (NISQ) devices where circuit depth must be kept extremely short to avoid decoherence.
A Survey of Quantum Computing for Finance <i>(Herman et al.)</i>	2022	The NISQ Shift: A metadata analysis detailing the industry's pivot away from fault-tolerant algorithms (like Shor's) toward practical, heuristic hybrid implementations like QAOA that offload heavy lifting	Provides the most up-to-date context on where the industry stands regarding commercial viability and error mitigation.

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		to classical optimizers.	
Portfolio Optimization with Quantum Annealing <i>(Hodson et al.)</i>	2019	The Integer Solution: Explores the limitations of QUBO when dealing with integer constraints (e.g., buying exactly 5 shares instead of just a binary yes/no). Solves this using binary expansion and mathematically scaling penalty functions.	Teaches the critical skill of penalty-function scaling, which is strictly required when building advanced QAOA models.
Dynamic Portfolio Optimization with Real Datasets Using Quantum Processors <i>(Mugel et al.)</i>	2020	Real-World Stress Testing: Validates theoretical QUBO models by back-testing quantum-selected portfolios against years of volatile, real-world historical market data to evaluate robustness during sudden market crashes.	Proves that theoretical QUBO mapping scales effectively to real-world, noisy data streams and market volatility.
Quantum Approximate Optimization Algorithm for Portfolio Optimization <i>(Slate et al.)</i>	2021	Depth vs. Returns: Systematically analyses how the number of layers (p-depth) in a QAOA circuit directly correlates to the financial Sharpe ratio of the generated portfolio, identifying where hardware noise wipes out theoretical gains.	Serves as a direct blueprint for building and benchmarking QAOA trading algorithms in Qiskit.
Financial Risk Management on a	2019	Quantum Amplitude Estimation: Shifts	Covers the risk-modeling side of finance;

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Quantum Computer (Woerner, Egger)		focus from asset selection to tail risk. Shows how encoding probability distributions into a quantum state evaluates Value at Risk (VaR) quadratically faster than classical Monte Carlo simulations.	optimization is functionally useless without accurate risk calculation.
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Title: A Quantum Approximate Optimization Algorithm (Farhi, Goldstone, Gutmann, 2014)

This foundational paper introduces the Quantum Approximate Optimization Algorithm (QAOA), a hybrid quantum-classical heuristic designed to solve combinatorial optimization problems on gate-model quantum computers. Moving away from the stringent hardware requirements of adiabatic quantum computing, the authors propose an alternating operator sequence consisting of a problem Hamiltonian (H_C), which encodes the objective function, and a mixing Hamiltonian (H_B), which drives state transitions to prevent convergence in local minima. By parameterizing these operators with angles γ and β , a classical optimizer iteratively updates the quantum circuit to locate the ground state of the target system. The authors rigorously demonstrate the algorithm's efficacy by applying it to the Max-Cut problem on 3-regular graphs, proving a guaranteed approximation ratio of 0.6924 at a circuit depth of $p = 1$, and establishing that the algorithm converges to the exact optimal solution as $p \rightarrow \infty$. This architecture forms the fundamental baseline for modern variational quantum algorithms utilized in quantitative finance.

Title: Quantum Computing for Finance: Overview and Prospects (Orús, Mugel, Lizaso, 2019)

This survey paper provides a foundational taxonomy for the application of quantum computing in the financial sector, identifying three primary pillars: optimization, machine learning, and derivative simulation. The authors bridge the gap between abstract quantum physics and institutional finance by mapping intractable market problems—specifically portfolio rebalancing and risk estimation—to quantum-compatible mathematical frameworks like Quadratic Unconstrained Binary Optimization (QUBO). A significant portion of the analysis is dedicated to the practical

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constraints of the NISQ (Noisy Intermediate-Scale Quantum) era, arguing that while full-scale fault-tolerant applications remain in the long term, hybrid quantum-classical algorithms like QAOA offer a viable, immediate path to achieving commercial quantum advantage. The paper concludes by defining the strategic roadmap for financial institutions to transition from classical simulation to quantum-enhanced decision-making, emphasizing that successful implementation requires expertise in both quantitative finance and quantum Hamiltonian mapping.

Title: Solving the Optimal Trading Trajectory Problem Using a Quantum Annealer (Rosenberg et al., 2016)

This research introduces the first practical application of quantum optimization to multi-period portfolio management, shifting focus from static asset selection to the temporal "trading trajectory." The authors formulate a high-dimensional objective function that balances expected returns, portfolio variance, and transaction costs across consecutive time steps. A critical methodological contribution is the implementation of Quadratic Unconstrained Binary Optimization (QUBO) to encode the multi-period constraints into a physical Ising model, utilizing Lagrange multipliers to penalize constraint violations. By executing this model on a quantum annealer, the study demonstrates that quantum systems can identify optimal trading paths that are computationally equivalent to, or more efficient than, classical heuristics. This work establishes the essential framework for mapping dynamic financial strategies onto quantum hardware and provides a rigorous methodology for penalty-function construction in constrained optimization.

Title: Quantum Optimization for Portfolio Management (Egger, Marecek, Woerner, 2020)

This paper provides a rigorous benchmarking of variational quantum algorithms—specifically QAOA and VQE—against the classical Markowitz Mean-Variance optimization framework. The authors formalize the mapping of the portfolio selection problem into a quadratic objective function, incorporating a risk-aversion coefficient to balance expected returns against asset covariance. A primary methodological contribution is the investigation of hardware noise impacts on circuit execution; the authors demonstrate that while small-scale portfolio instances yield optimal asset allocations, current-generation superconducting hardware suffers from significant depolarizing and readout noise that flattens the energy landscape and obscures the global minimum. The paper concludes that effective quantum portfolio optimization in the NISQ era necessitates both robust error mitigation strategies and pre-processing techniques to reduce the dimensionality of the asset pool before quantum execution.

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Title: Warm-Starting Quantum Optimization (Egger, Marecek, Woerner, 2021)

This research introduces the "Warm-Starting" methodology for variational quantum algorithms, addressing the fundamental trade-off between circuit depth and hardware decoherence. The authors demonstrate that initializing the QAOA circuit in a uniform superposition is suboptimal for complex financial instances; instead, they propose leveraging classical relaxation algorithms to generate an informed probability distribution, which is then injected into the quantum state as the initial starting point. This approach effectively circumvents the "Barren Plateau" phenomenon by constraining the algorithm's search space to the neighborhood of a classically-derived estimate. The methodological contribution is twofold: it significantly reduces the quantum circuit depth (p) required to achieve convergence—thereby enhancing viability on current-generation noisy (NISQ) hardware—and provides a framework for integrating classical heuristics with quantum processing to refine portfolio allocations beyond the limitations of purely classical solvers.

Title: A Survey of Quantum Computing for Finance (Herman et al., 2022)

This survey paper provides a comprehensive metadata analysis of the industry's progression toward commercial quantum advantage, documenting a pivotal shift from theoretical fault-tolerant algorithms to practical, heuristic variational approaches. The authors argue that while algorithms like Quantum Phase Estimation offer definitive exponential speedups, their hardware requirements remain prohibitive for near-term financial applications. Consequently, the paper emphasizes the dominance of hybrid quantum-classical frameworks, such as QAOA and VQE, which leverage classical pre-processing to reduce dimensionality before executing the combinatorial core on NISQ hardware. The research explicitly identifies critical industry bottlenecks, including data-loading latency (the QRAM problem) and hardware decoherence, while advocating for robust error-mitigation protocols like Zero-Noise Extrapolation as essential prerequisites for practical financial deployment. This survey serves as a definitive benchmark for evaluating the current technological maturity of quantum optimization relative to traditional classical heuristic solvers.

Title: Portfolio Optimization with Quantum Annealing (Hodson et al., 2019)

This research addresses the practical limitations of mapping complex, non-binary financial portfolios onto quantum hardware, specifically tackling the encoding of integer asset holdings into binary-constrained QUBO formulations. The authors implement two primary mapping strategies: unary encoding, which maximizes qubit efficiency for small integer ranges, and binary expansion, which scales effectively for larger asset portfolios.

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A significant methodological contribution is the derivation of a penalty-scaling protocol; the authors define the optimal Lagrange multiplier (λ) as a function of the portfolio's return range, ensuring that budget constraints are enforced without degrading the resolution of the objective function. The study provides a blueprint for the "Hamiltonian engineering" required in real-world deployments, demonstrating that the success of quantum optimization is contingent upon the meticulous balancing of energy penalties against target returns, a prerequisite for scaling to institutional-grade portfolios.

Title: Financial Risk Management on a Quantum Computer (Woerner & Egger, 2019)

This landmark study introduces a quantum-native framework for risk management, specifically targeting the calculation of Value at Risk (VaR) and Conditional Value at Risk (CVaR). The authors propose replacing conventional Monte Carlo simulations with Quantum Amplitude Estimation (QAE), a quantum algorithm derived from Grover's amplitude amplification that achieves a quadratic speedup—reducing sample complexity from $O(1/\epsilon^2)$ to $O(1/\epsilon)$. The methodological framework involves three stages: the probabilistic encoding of market uncertainty into a quantum register, the construction of an oracle function to identify tail-risk outcomes, and the use of amplitude amplification to estimate the probability of those events with superior precision. The research validates that this quantum approach not only improves the computational efficiency of derivative pricing but also enables real-time risk assessment for portfolios with complex, non-linear dependencies, providing a concrete mathematical basis for the adoption of quantum hardware in regulatory and institutional risk management.

Title: Dynamic Portfolio Optimization with Real Datasets Using Quantum Processors (Mugel et al., 2020)

This empirical study evaluates the viability of quantum optimization models by subjecting them to backtesting against longitudinal historical market data, moving beyond the toy-problem constraints of previous literature. The methodology employs a rolling-window optimization framework to assess portfolio robustness during periods of both stability and high market volatility. A key technical finding is the resilience of QUBO-based quantum models when faced with unstable covariance matrices during market crashes; the study demonstrates that quantum optimization effectively identifies uncorrelated asset classes that classical matrix-inversion methods often fail to isolate. By integrating classical noise-reduction techniques (Random Matrix Theory) with quantum heuristic solvers, the authors provide a scalable blueprint for financial institutions to transition from theoretical portfolio construction to active, dynamic asset

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management, proving that quantum-derived portfolios can remain competitive with, and occasionally outperform, classical mean-variance benchmarks in live market conditions.

Title: QAOA for Portfolio Optimization (Slate, O'Malley, & Jordan, 2021)

This research provides a rigorous technical analysis of the performance metrics of QAOA applied to portfolio selection, specifically investigating the correlation between circuit depth (p) and the resulting financial Sharpe ratio. The authors identify a non-linear relationship where increasing circuit depth yields logarithmic improvements in portfolio accuracy up to a hardware-specific threshold, beyond which the accumulation of gate-based noise and decoherence outweighs the mathematical precision of the algorithm. By defining this "Goldilocks Zone" of optimal circuit depth, the study provides a practical methodology for hardware-efficient benchmarking; the authors conclude that institutional-grade quantum finance strategies must balance the theoretical benefits of higher p -depths against the reality of current-generation hardware noise. This paper serves as the final technical blueprint for deploying and scaling quantum optimization algorithms in production financial environments.